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香港考試及評核局
HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

2013年香港中學文憑
HONG KONG DIPLOMA OF SECONDARY EDUCATION 2013

CHEMISTRY PAPER 1 & COMBINED SCIENCE (CHEMISTRY) SECTION B

MARKING SCHEME

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INSTRUCTIONS TO MARKERS

1. In order to maintain a uniform standard in marking, markers should adhere to the marking scheme agreed at the markers' meeting.

2. The marking scheme may not exhaust all possible answers for each question. Markers should exercise their professional discretion and judgment in accepting alternative answers that are not in the marking scheme but are correct and well reasoned.

3. The following symbols are used:

/	A single slash indicates an acceptable alternative within an answer.
*	Step-mark (for questions involving calculations)
†	Correct spelling required

4. In questions asking for a specified number of reasons or examples etc. and a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, and if a candidate gives three answers, only the first two should be marked.

5. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

6. Award zero marks for answers which are contradictory.

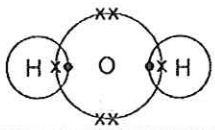
7. Chemical equations should be balanced except those in reaction schemes for organic synthesis. For energetics, the chemical equations given should include the correct state symbols of the chemical species involved.

8. In the question paper, questions which assess candidates' communication skills are marked with an asterisk (*). For these questions, the mark for effective communication (1 mark per question) will be awarded if candidates can produce answers which are easily understandable. No marks for effective communication will be awarded if the answers produced by candidates contain a lot of irrelevant materials and/or wrong concepts in chemistry.

Part I

Marks

1. (a)



1

- Must show 'H' & 'O'
- Accept electrons NOT drawn on circles (shells)
- Accept any symbols to represent electrons
- If the inner shell is shown, total number of shells and electrons should be correct

(b) (i) Water boils at about 100°C, but the salts in sea water are non-volatile / boiling point of water is lower than that of salt. 1

The steam (water vapour) formed condenses on the cold surface of the condenser / cool down to give liquid water (the distillate). 1

(ii) To prevent bumping / to prevent frothing / splash / overflow due to overheating of water. 1

To ensure smooth boiling. (1)

(c) The attraction between water molecules is predominately hydrogen bond. 1

Hydrogen bond is directional. In ice, the H_2O molecules have a tetrahedral arrangement / are packed in an open structure. 1

In liquid water, the H_2O molecules have relative motion and this leads to the collapse of the open structure. The molecules become more closely packed. ∴ liquid water has higher density than ice. 1

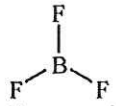
For CS:

(i) crystallisation / recrystallisation 1

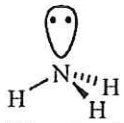
(ii) Conduct a flame test / correct outline of the procedure. 1

A golden yellow flame shows the presence of sodium. 1

Marks

2. (a)  1

(Accept other correct representations of the trigonal planar structure.)

 1

(Showing the lone pair is NOT a must.)

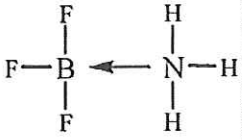
(b) BF_3 is a non-polar molecule. The three polar B–F bonds are symmetrically arranged on the same plane / dipole moments cancel out / net dipole moment is 0. 1

NH_3 is a polar molecule. The molecule has a lone pair in its outermost shell and thus the three polar N–H bonds are not on the same plane / dipole moments cannot cancel out / net dipole moment is not 0. 1

(Only award marks to the explanations.)

(c) In BF_3, there are three (bond) electron pairs / there is a vacant site / 6 electrons only / electron deficient in the outermost shell of the B atom. 1

By accepting the lone pair of electrons from the nitrogen atom of NH_3 / forming dative bond with N, boron attains the stable electronic configuration of neon (a noble gas) / diagram below 1



(Showing the lone pair is NOT a must) (1)

2. *For CS:*

(a) (i) KNO_3 is added to react with sodium which is highly reactive / corrosive / flammable / strongly reducing. 1

(ii) Oxidising agent: the oxidation number of N decreases from +5 (in KNO_3) to 0 (in N_2). 1

Oxidation number of Na increases from 0 (in Na) to +1 (in Na^+), thus KNO_3 is the oxidising agent. (1)

(b) (i) $\frac{0.01 - 0.005}{10}$ 1

$= 0.0005$ mol dm^{-3} s^{-1} (5.0×10^{-4}mol dm^{-3} s^{-1})

(Accept the unit M^{-1}.)

(Accept answers such as -0.0005 mol dm^{-3} s^{-1}, 0.003 mol dm^{-3} min^{-1}, 1.8 mol dm^{-3} h^{-1}.)

(ii) Determine the slope of the tangent of the curve at t = 10 s. 1

Marks

3. (a) Atom ratio $\frac{\text{C}}{\frac{2.64}{44}} : \frac{\text{H}}{2\times\frac{1.08}{18}} : \frac{\text{O}}{\frac{0.48}{16}}$ 1*

$= 2 : 4 : 1$

Empirical formula is C_2H_4O

Molecular formula is $(C_2H_4O)_n$

$n\times(12\times2 + 1\times4 + 16\times1) = 88.0$ 1*

$n = 2$

molecular formula of W is $C_4H_8O_2$ 1

Alternative method:

No. of C atoms in one molecule of W

$= \frac{2.64}{44}\times\frac{88}{1.32}$ (1*)

$= 4$

No. of H atoms in one molecule of W

$= \frac{1.08}{18}\times\frac{88}{1.32}\times2$

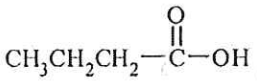
$= 8$

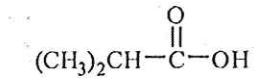
No. of O atoms $= \frac{88-12\times4-8}{16}$ (1*)

$= 2$

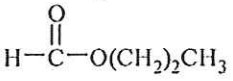
molecular formula of W is $C_4H_8O_2$ (1)

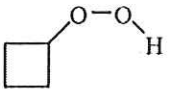
(Accept other correct methods.) (Steps and answer are to be marked separately)

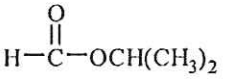
(b)  1

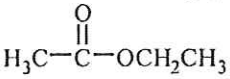
 1

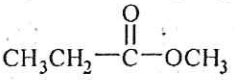
Also accept the following structures:

 (1)



or  (1)

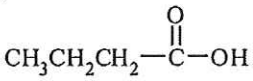
or  (1)

or  (1)

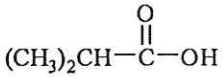
Marks

For CS:

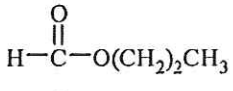
(b) (i) One mark for the structure of a carboxylic acid: 1

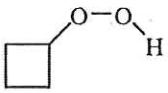


(1)

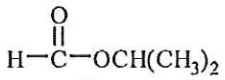
or  (1)

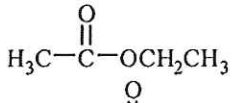
One mark for the structure of an ester: 1

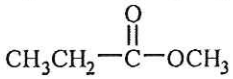




(1)

or  (1)

or  (1)

or  (1)

(b) (ii)
Treat the two compounds separately with Na_2CO_3 / $NaHCO_3$ / appropriate metals e.g. Mg. 1
Only the carboxylic acid will give effervescence. 1
(Accept other chemical tests that can distinguish the two suggested compounds, e.g. titration.)

4. (a) †Ethanedioic acid 1
(Do not accept "oxalic acid")

(b) $H_2C_2O_4(aq) \rightleftharpoons C_2O_4^{2-}(aq) + 2H^+(aq)$ / $H_2C_2O_4(aq) \rightleftharpoons HC_2O_4^-(aq) + H^+(aq)$ 1
$H_2C_2O_4$ is a weak acid. It undergoes incomplete ionisation in water. (1)
As pH = $-\log_{10}[H^+(aq)]$ and $[H_3O^+(aq)]$ in 0.05 M $H_2C_2O_4(aq)$ is less than 0.1 M, its pH is thus greater than 1. 1

(c) NaOH(s) is deliquescent / hygroscopic / absorbs water from the atmosphere. 1
Or, NaOH(s) reacts with $CO_2(g)$ in the atmosphere. (1)
∴ The mass of NaOH(s) cannot be accurately determined by weighing.

(d) (i) From colourless to pink / purple / red 1

(ii) concentration of NaOH(aq)
$= \frac{0.05 \times 25 \times 2}{17.20}$ (If the mole ratio of reactants is correct 1:2, award step mark) 1*
$= 0.145$ mol dm^{-3} (M) (Unit is optional) 1
(Also accept 0.15 mol dm^{-3})

(e) (i) Rinsing the conical flask with $H_2C_2O_4(aq)$: Some $H^+(aq)$ ions / acid / $H_2C_2O_4(aq)$ remain in the flask, and more alkali (as revealed from the burette reading) than actually required is used to reach the titration end-point. 1
(Do not accept the concentration of $H^+(aq)$ increases.)

(ii) NaOH(aq) clinging onto the stem of funnel may fall into the burette. The volume of alkali used (as revealed from the burette reading) is smaller than what is expected. 1

†: correct spelling
* step mark

5. (a) (i) $(26.6 - 19.8) = 6.8°C$ (The unit must be correct) 1
(Acceptable range : 6.7 to 6.9°C; also accept answer with a negative sign)

(ii) Heat absorbed $= mc\Delta T$
$= \underline{21.8 \times 4.3 \times 6.8}$ 1*
$= 637.4$ J (The unit "J" can be ignored) (1*)

No. of moles of $NH_4NO_3(s)$ used $= \underline{2.0 / 80}$ 1*
$= 0.025$ (1*)

$$\Delta H = +\frac{637.4}{0.025}$$
$= +25.5$ kJ mol^{-1} (The answer must have the correct unit and the "+" sign) 1
(Accept +25500 J mol^{-1}) (Accept: +24.7, +25.1 — 25.9 kJ mol^{-1})
(Accept answer round-off to 2 significant figures : +25.1 kJ mol^{-1} → 25 kJ mol^{-1}; 25.9 kJ mol^{-1} → 26 kJ mol^{-1})
(Accept if the candidate use 6.6°C for the calculation, the answer will be +24.7 kJ mol^{-1}.)

(b) Place $NH_4NO_3(s)$ in a desiccator/sealed container containing drying agent (anhydrous $CaCl_2(s)$, $CaO(s)$, $MgSO_4(s)$, etc.) / silica gel. 1
(Accept either stored in desiccator/sealed container, or stored in the presence of a drying agent with a proper example. For $CaCl_2(s)$, candidate should mention it is anhydrous.)
(Also accept: Place NH_4NO_3 in a room/an area with low humidity control)
(Not accept: mixing NH_4NO_3 with the drying agent; Not accept NaOH(s) and conc. H_2SO_4 as the drying agent)

6. • Cracking of naphtha gives a mixture of hydrocarbons which include propene. 1
• Fractional distillation of the gaseous products can separate propene from other hydrocarbons. 1
• Polymerisation of propene at elevated temperatures / > 45℃ / high pressure / > 5 atm / in the presence of a suitable catalyst /Ziegler-Natta catalyst gives polypropene (Polymerisation + any 1 condition). 1
(Polymerisation of propene can be described in the form of a chemical equation.)

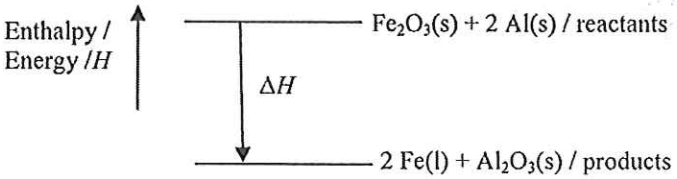
• Communication mark 1
(chemical knowledge = 0 to 1, communication mark = 0
chemical knowledge = 2 to 3, communication mark = 0 or 1
incomplete answer / difficult to understand, communication mark = 0)

* step mark

Marks

7. (a) (i) $Fe_2O_3(s) + 2\ Al(s) \rightarrow 2\ Fe(l) + Al_2O_3(s)$ 1
(State symbols are not required)

(ii)



1

(1. State symbols are not required.
2. The enthalpy level for the products must be lower than that of the reactants. The drawing should either show the proper "y-axis", or have the "ΔH" symbol shown in the drawing.
3. If the energy levels are labeled with chemical formula of reactants and products, the formula should be correct. Coefficients of the chemical substances can be ignored.)

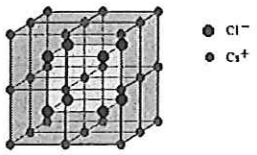
(b) Copper is less reactive than iron / copper has a lower affinity for oxygen than iron. / Copper is a weaker reducing agent than iron. / Copper is lower than iron in the chemical reactivity series / electrochemical series. 1
∴ Cu(s) cannot reduce $Fe_2O_3(s)$.

(c) (i) Aluminium is more expensive than iron. / Using aluminium to extract iron is costly. 1

(ii) coke / carbon / charcoal / carbon monoxide / CO 1
(Not accept coal or H_2)

8. (a) $2\ Cs + Cl_2 \rightarrow 2\ CsCl$ (State symbols are not required) 1

(b) (i) 1



(The drawing should either show the correct labels for Cs^+ and Cl^-, or show clearly there are two types of ions in the lattice with correct relative positions.)
(Accept other reasonable representations.)

(ii) CsCl contains Cs^+ / cations and Cl^- / anions. / In CsCl , ions are strongly held by ionic bond. 1
Relative movement of the ions can bring ions of the same charge close to each other, and will result in repulsion. ∴ CsCl(s) is brittle. 1

(c) Cs(s) is more reactive than Na(s). The reactivity of Group 1 metals increases down the group./ 1
The electron in the outermost shell (valence electron) of Cs is weakly bounded by the nucleus as compared with that of Na. / (1)
Cs atom loses its outermost shell electron more easily than Na atom. / (1)
Cs atom loses its electron more easily than Na atom because Cs has more electron shells than Na / the size of Cs atom is larger than that of Na / the atomic radius of Cs is larger than that of Na./ (1)
Both Cs and Na are Group I metals, and the size of Cs atom is larger than that of Na. (1)

Marks

9. (a) Brown colour / yellow colour is observed. 1
Due to the high concentration of KI in the solution, I^- ions are preferentially discharged to give I_2 / discharged instead of OH^- to give I_2 which dissolves in KI(aq) to give brown I_3^- ions. 1
(Note - minimum requirement: concentration effect + preferentially discharge of I^- / high concentration of KI + discharge of I^-)

(b) (i) H^+ is discharged / reduced to H_2 at electrode B. 1
$2\ H^+(aq) + 2e^- \rightarrow H_2(g)$ (1)
Depletion of H^+ ions makes $[OH^-(aq)] >> [H^+(aq)]$ / The amount of OH^- ions increases at electrode B as H^+ is being consumed. 1
∴ Universal indicator turns blue under alkaline conditions.

(ii) Accept both 'yes' and 'no' answers. Award 1 mark for a reasonable explanation. 1
'No': B is the negative electrode. Copper will not lose electrons to give Cu^{2+} at the negative electrode / Copper (Cu) cannot undergo reduction at the negative electrode. / Copper will not take part in chemical changes and will act only as the electrode. (1)
'Yes': Copper and carbon have different electrical conductivity. Therefore the solution near electrode B turns blue more quickly / The current in the external circuit changes. (1)

10. (a)  1

(b) electrode D: $H_2(g) + 2\ OH^-(aq) \rightarrow 2\ H_2O(\ell) + 2e^-$ / $2\ H_2(g) + 4\ OH^-(aq) \rightarrow 4\ H_2O(\ell) + 4e^-$ 1

electrode E: $O_2(g) + 2\ H_2O(\ell) + 4e^- \rightarrow 4OH^-(aq)$ 1

(c) (i) Accept both 'agree' and 'disagree' answers. Award 1 mark for a sound argument. 1
Agree: The hydrogen can be obtained from renewable source (with one proper example) (1)
(E.g. Electrolysis of water using the electricity generated from hydropower / reforming of CH_4 obtained from animal manure.)
Disagree: The hydrogen gas used is produced from fossil fuel such as steam reforming of natural gas. (1)
Disagree: (Electrical) energy is consumed in the production of hydrgoen (from water). (1)
(NOT Accept the answer is yes, because the hydrogen can be obtained from the electrolysis of water, and so the fuel cells do not consume fossil fuel.)

(ii) Agree: Only water is produced from the hydrogen-oxygen fuel cells / 1
Agree: No CO_2 / SO_2 / NO_x / CO / unburnt hydrocarbon in the exhaust (1)

Part II

Marks

11. (a) The airbag has to be inflated instantly when a car crash occurs.
Fine powder can greatly increase the reaction rate / can give a fast reaction by providing a (very) large surface area for a reaction involving solid reactants. 1

(b) No. of mole of N_2 produced from the decomposition of NaN_3 (reaction 1)
$= \frac{100}{65} \times \frac{3}{2} = 2.31$ 1*

No. of mole of N_2 produced from the reaction of Na with KNO_3 (reaction 2):

No. of mole of Na produced = $\frac{100}{65} = 1.54$

(No. of mole of KNO_3 = $\frac{200}{101.1} = 1.98$

Since 5 mol of Na react with 1 mol of KNO_3, KNO_3 is in excess.)

No. of mole of N_2 produced from reaction 2 = $\frac{100}{65} \times \frac{1}{10} = 0.154$ 1*

Volume of gas produced = $(2.31 + 0.154) \times 24 = 59.1$ dm^3 (Correct unit is required) 1
(Accept: 58.8 – 59.2 dm^3)

(c) KNO_3 is added to react with sodium which is (highly) reactive / corrosive / flammable / strongly reducing. 1

(d) (i) $\frac{0.01 - 0.005}{10}$
$= 0.0005$ mol dm^{-3} s^{-1} (5.0×10^{-4} mol dm^{-3} s^{-1}) (Correct unit is required) 1
(Accept answer with a negative "–" sign)
(Accept 0.0005 M s^{-1} / 0.03 mol dm^{-3} min^{-1} / 1.8 mol dm^{-3} hr^{-1})

(ii) Determine the slope of the tangent of curve at t = 10 s. 1

* step mark

Marks

12. (a) Reaction quotient = $\frac{(0.04)}{(0.05)(0.02)}$ $mol^{-1}\ dm^3$

$= 40\ mol^{-1}\ dm^3$ (Accept answer without unit) (NOT accept wrong unit) 1

∵ Reaction quotient > K_c

∴ Backward reaction rate is greater than the forward reaction rate. 1

(Remark: If the process of deduction is correct, 1 mark will be awarded even though the candidate failed to obtain the correct result for "Reaction quotient".
For example, if the candidate incorrectly obtained "6" for the reaction quotient, and deduced the forward reaction rate is faster because the reaction quotient is smaller than K_c, award 1 mark)

(b) At equilibrium, the concentrations are:

$[PCl_5(g)] = (0.04 - y)\ mol\ dm^{-3}$
$[PCl_3(g)] = (0.05 + y)\ mol\ dm^{-3}$
$[Cl_2(g)] = (0.02 + y)\ mol\ dm^{-3}$

Also accept :
$[PCl_5(g)] = (0.04 + z)\ mol\ dm^{-3}$
$[PCl_3(g)] = (0.05 - z)\ mol\ dm^{-3}$
$[Cl_2(g)] = (0.02 - z)\ mol\ dm^{-3}$

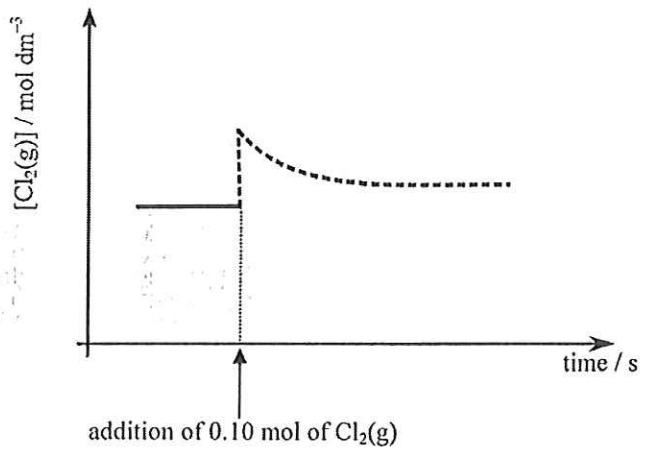
$\frac{(0.04 - y)}{(0.05 + y)(0.02 + y)} = 25$

Also accept: $\frac{(0.04 + z)}{(0.05 - z)(0.02 - z)} = 25$ 1*

Solving equation, $y = 0.0052$

$[Cl_2(g)]_{eqm} = (0.02 + 0.0052)\ mol\ dm^{-3} = 0.0252\ mol\ dm^{-3}$ ($0.025\ mol\ dm^{-3}$) 1

(c)



1

(Remarks: The final equilibrium level of $[Cl_2]$ should lie between the original level and the level when 0.1 mol of Cl_2 was just added.)

13. Nitrogen < Lithium < Beryllium < Carbon (graphite) 1

N_2 has the lowest melting point as it has a simple molecular structure, weak van der Waals' forces / intermolecular forces need to be overcome. 1

Both Li and Be have (metallic structure), metallic bond in Li is weaker than that in Be ∴ Li < Be in melting points. 1

C has the highest melting point as it has a giant covalent structure, (large amount of energy is needed to) break (millions of) strong covalent bonds between atoms in melting 1

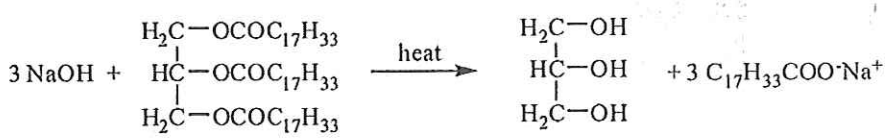
Effective communication 1

(chemical knowledge = 0 , 1 or 2, communication mark = 0
chemical knowledge = 3 or 4, communication mark = 0 or 1
incomplete answer / difficult to understand, communication mark = 0)

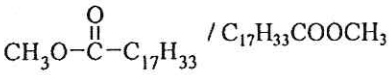
Marks

14. (a) H_2, Ni / Pd / Pt 1

(b) (i)



1

(ii)  / $C_{17}H_{33}COOCH_3$ 1

(c) G has a smaller relative molecular mass than F, so G can be vaporized more easily than F. 1
G has a smaller relative molecular mass than F, so G has a lower boiling point than F. (1)
G has a smaller relative molecular mass than F, and so the molecular size of G is smaller than that of F. The intermolecular attraction / van der Waals' forces between G are weaker than that between F, G can be vaporized more easily than F. (1)

G burns more completely/ more easily than F. 1

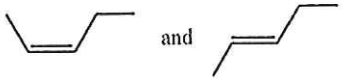
15. (a) Correct chemical reagent 1
Correct observations with comparison between the tests on X and Y 1

Possible tests and the corresponding observations:

$Cr_2O_7^{2-}$ / H^+	Observations: X – no change; Y – from orange to green
MnO_4^- / H^+	Observations: X – no change; Y – from purple to colourless
MnO_4^- / OH^-	Observations: X – no change; Y– formation of brown ppt.
2,4-DNP	Observations: X – formation of orange ppt.; Y– no change
CH_3COOH / H^+ / heat	Observations: X – no change; Y– fruity smell substance formed

2,4-DNP = 2,4-dinitrophenylhydrazine
(Accept other chemical tests that can distinguish a ketone from an alkanol, e.g. Na / PCl_5, etc.)

(b) $LiAlH_4$ / $NaBH_4$ 1

(c)  and 1

(The drawings should show the positions and cis/trans-geometry of the C=C double bonds clearly and correctly. It is not necessary to have the bond angles drawn exactly as 120°)

Geometrical (isomerism) / cis/trans- (isomerism) 1

(d)  / $CH_3CHClCH_2CH_2CH_3$ 1

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香港考試及評核局
HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

2013年香港中學文憑
HONG KONG DIPLOMA OF SECONDARY EDUCATION 2013

CHEMISTRY PAPER 2

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INSTRUCTIONS TO MARKERS

1. In order to maintain a uniform standard in marking, markers should adhere to the marking scheme agreed at the markers' meeting.

2. The marking scheme may not exhaust all possible answers for each question. Markers should exercise their professional discretion and judgment in accepting alternative answers that are not in the marking scheme but are correct and well reasoned.

3. The following symbols are used:

/	A single slash indicates an acceptable alternative within an answer.
*	Step-mark (for questions involving calculations)
†	Correct spelling required

4. In questions asking for a specified number of reasons or examples etc. and a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, and if a candidate gives three answers, only the first two should be marked.

5. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

6. Award zero marks for answers which are contradictory.

7. Chemical equations should be balanced except those in reaction schemes for organic synthesis. For energetics, the chemical equations given should include the correct state symbols of the chemical species involved.

Marks

1. (a)(i)(1) *x* --- (molecular kinetic) energy / K.E. 1

y --- fraction/percentage / number / amount of molecules / particles 1

(2) (From the sketch), an increase in temperature (from T_1 to T_2) will increase the (average kinetic) energy of the molecules. 1

This will increase the number of collisions / increase the collision frequency / result in greater effective collisions. 1

(From the sketch,) there is a larger portion of molecules will have kinetic energy greater than E_a. 1

(ii) $\log k = \text{constant} - \frac{E_a}{2.3RT}$ 1

OR $k = Ae^{-Ea/RT}$ (1)

OR $\ln k = \ln A - Ea/RT$ BUT not accept In as ln (1)

Slope of curve $= -\frac{E_a}{2.3R}$ 1*

$= -1.73 \times 10^3$ (Range: –1.67 — –2.05) (1*)

$E_a = 1.73 \times 10^3 \times (8.31 \text{ J K}^{-1} \text{ mol}^{-1}) \times 2.3$

$= 33.1 \text{ kJ mol}^{-1}$ OR 33 100 J mol^{-1} Accept mol^{-} as mol^{-1} 1

(Acceptable range: 32 to 39 kJ mol^{-1})

Accept determination based on equation:

$(\log k_1 - \log k_2)/(1/T_1 - 1/T_2) = -Ea/2.3R$

(b) (i) A catalyst provides an alternative pathway with lower activation energy. 1

(ii) Concentrated H_2SO_4

Dilute H_2SO_4 contains a lot of H_2O. Water can shift the equilibrium position to the left / cause hydrolysis of ester (eugenol benzoate) / reactant side, and thus lowers the yield of the product. 1

Also accept: conc. H_2SO_4 is a dehydrating agent / removes water from the product side of the reaction and will shift the equilibrium position to the right / product side. (1)

(iii) Homogeneous catalyst --- H_2SO_4 is readily available / commonly found 1

Heterogeneous catalyst --- can be reused / easily be regenerated / can be separated easily 1

* step mark

Marks

(c) (i) Chlorine can be produced industrially by electrolysis of brine /concentrated sodium chloride solution using flowing mercury cell / diaphragm cell / membrane cell. 1

Cl^- (aq) ions are discharged at the anode to give Cl_2(g). 1

Anode: $2\ Cl^-(aq) \rightarrow Cl_2(g) + 2\ e^-$ (1)

Diaphragm cell / membrane cell

H^+(aq) ions are discharged at the cathode. With the removal of Cl^-(aq) ions and H^+(aq) ions, the resultant electrolytic solution contains NaOH(aq) in high concentration. 1

[If Cl^- (aq) ions mentioned above, accept only mention H^+(aq).]

Or, overall reaction: $2\ NaCl + 2\ H_2O \rightarrow Cl_2 + H_2 + 2\ NaOH$ (1)

OR

Flowing mercury cell

Sodium amalgam produced at cathode reacts with water forming NaOH(aq). (1)

Or, $2Na + 2H_2O \rightarrow 2NaOH + H_2$ (1)

(ii) Method 2: 1

$$\text{Atom economy} = \frac{58}{76} = 76.32\%$$

[Accept also 76, 76.3, 76.316]

[If the fraction given, the numbers therein must be correct.]

(iii) Method 2 is greener. 2

Any two of the following:

--- It has a higher atom economy.

--- It produces less waste (less side products) and less treatment is required / requires less steps.

--- Less hazardous chemicals are used such as method 1 uses (more) toxic Cl_2 / Method 1 uses more corrosive NaOH / Method 1 uses more toxic Cl_2

(Accept any reasonable answer based on the principles of green chemistry and note for comparative sense.)

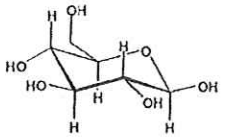
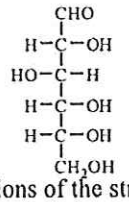
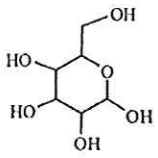
(iv) The calculation of atom economy is based on a 100% completed reaction. / complete conversion of reactants to products / Give definition of atom economy / high atom economy ⇒ less side product 1

Most reactions do not go to completion / There is loss during handling. / The yield is related to the extent of reaction (percentage conversion). 1

Thus a reaction with high atom economy does not necessarily have a high yield.

Marks

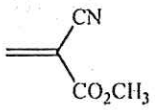
2. (a)(i)(1) A condensation polymer is a polymer which when formed from its monomers involves the elimination of small (water) molecules (Do not accept HCl, NH_3). 1

(2)  or  or  1
(Accept other correct representations of the structure of glucose.)
(Ignore the positions of –OH groups in the molecule.)

(ii) Molecules of cellulose may contain various number of glucose molecules joined together. 1

(iii) Glucose is very soluble in water while cellulose is insoluble in water. 1
A glucose molecule has 5 –OH / hydroxyl groups which can interact with water molecules strongly via hydrogen bonds. 1
The –OH / hydroxyl groups of a cellulose molecule form hydrogen bonds with other cellulose molecules, therefore they are less available to form hydrogen bonds with water molecules. 1

(b) (i) The solid becomes soften / melt upon heating. 1
The heat energy absorbed helps the polymeric molecules to overcome the intermolecular attraction and the molecules can have relative translational motion. 1
At vey high temperature, the glue chars / burns. 1
Sufficient heat energy causes the polymeric molecules to undergo bond breaking / decomposition / reaction with oxygen, forming carbon and other compounds. (1)

(ii)(1)  1

(2) Both propanone and poly(methyl 2-cyanoacrylate) are polar organic molecules. 1
The intermolecular attraction in the two compounds is of the same type (polar attraction). Thus, propanone can dissolve poly(methyl 2-cyanoacrylate). 1
('Like dissolves like')

(iii) Methyl cellulose is more readily degraded in the environment.
Methyl cellulose is produced from cellulose / glucose, a natural material, which is more biodegradable / can be degraded by actions of bacteria or enzymes. 1
Poly(methyl 2-cyanoacrylate) contains long carbon chains which are not readily decomposed. 1

(c) (i) In both nematic phase and smectic phase, the molecules are directionally ordered along one direction. 1
In the smectic phase, the molecules are positionally aligned with each other. In the nematic phase, they are not aligned. 1

(ii) Molecule A exhibits the cholesteric phase as it is chiral and only chiral compounds can exhibit the cholesteric phase. 1

(iii) The compound solidifies at a very low temperature. 1

(iv) Liquid crystals displays need a backlight source, but OLED displays do not. 1
Liquid crystals act as optical filters to filter off the light from the backlight to give dark spots. Only a small fraction of light passes through them and makes up the images. 1

Marks

3. (a) Use separating funnel to remove water (the lower liquid layer) from the mixture. 1
The remaining upper/organic layer (remaining in the funnel) is hex-1-ene and octane. 1
Carry out fractional distillation / distillation on the upper layer: 1
The first distillate collected is hex-1-ene. The second distillate collected is octane. 1
If candidate reversed the steps (i.e. carry (fractional) distillation before separating funnel), deduct 1 mark.
If candidates only drew diagrams without any explanation, deduct 1 mark.

(b) (i) Any ONE of the following: 1
--- a significant observable change can easily be detected with the complete consumption of the reagent in the conical flask, i.e. availability of suitable indicator
--- the reaction must be significantly complete / it is not a reversible reaction
--- the rate of reaction must be fast enough to be practical
(Accept other correct answer.)

(ii) SO_2 easily escapes from solution / I_2 is volatile. 1

(iii) from colourless to blue-black / blue / dark blue 1
[i.e. must have the word "blue";
"to" can be replaced by →]

(iv) No. of mole of I_2 used $= 0.00412 \times 10.50 \times 10^{-3}$ 1
$= 4.33 \times 10^{-5}$ (1)
No. of mole of SO_2 = No .of mole of I_2 1*
$= 4.33 \times 10^{-5}$ (1*)

Mass of SO_2 in the wine sample $= 4.33 \times 10^{-5} \times 64.1$ 1*
$= 2.77 \times 10^{-3}$ g (1*)
$= 2.77$ mg (1*)

Concentration of SO_2 in the wine sample $= \dfrac{2.77}{0.025} = 111$ (mg dm^{-3}) 1
[Range: 110 – 112]

(v) This method cannot be used as the intense red colour of red wine may mask the colour of iodine / iodine-starch complex which leads to a difficulty in the end-point detection. 1
Or
This method can be used if the colour of red wine can be removed. (1)

(c) (i) $\dfrac{2.3}{2.3+1.9+2.9} = \dfrac{2.3}{7.1} = 0.32$ [Accept also 0.324 and 0.3239 BUT NOT 0.3] 1
[If the fraction given, the numbers therein must be correct.]

(ii) (β)-carotene because 1
lycopene has a smaller R_f value / lycopene moves slower /
lycopene takes a longer time to reach the bottom of the column.
OR argue in terms of (β)-carotene

* step mark

Marks

(iii) Method: colorimetry / (use of) colorimeter / use of spectrophotometer 1

Measurement: absorbance / colour intensity / light intensity / transmittance 1

(iv) Compound W is lycopene. 1

The absence of (absorption) peaks at around 2070 – 2250 (cm^{-1}) suggesting it does not contain C≡C groups / ruling out the possibility of compound X. 1

The absence of (absorption) peaks at around 2200 – 2280 (cm^{-1}) suggesting it does not contain C≡N groups / ruling out the possibility of compound Y. 1
OR
The presence of (absorption) peaks at around 1630 (cm^{-1}) (1600 – 1680 cm^{-1}) (shows the presence of C=C/C-C double bonds) ruling out the possibility of compound Y. (1)

(Note: 2 marks for the absence of (absorption) peaks at around 2070 – 2280 (cm^{-1}), suggesting it does not contain C≡C and C≡N groups / ruling out (the possibility of compounds) X and Y.)

The absence of sharp (absorption) peaks at around 3350 – 3500 (cm^{-1}) suggesting it does not contain $-NH_2$ groups / ruling out the possibility of compound Z. 1

Compound W is a hydrocarbon which contains only C-H, C-C and C=C bonds. This agrees with the feature of the spectrum as there are small peaks at around 1630 cm^{-1}, showing the presence of C-C double bonds.